A3-PS005-TC020

Explainable BGP Origin and Subprefix Anomaly Monitor

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Problem Description

BGP accepts whatever it is told, and RPKI origin validation covers only part of the problem.

- **No origin check:** any AS can announce any prefix and neighbours usually accept it.
- **Subprefix hijacks:** routers prefer the longest prefix, so a /24 announcement pulls traffic from the real /22 (YouTube 2008, Amazon Route 53 2018).
- **Gaps in ROV:** no ROA means NotFound, and the route is still accepted. A loose maxLength lets a forged-origin subprefix pass as Valid; about 84% of maxLength prefixes were exposed in 2017 [4].
- **False alarms:** multi-homing, provider changes and DDoS mitigation look like hijacks from the outside.
- **Timing:** each update must be checked against the ROAs valid when it was sent.

What PS-005 asks for

- Flag unexpected origins and subprefix announcements, with evidence for every alert.
- Rule-based baseline first, then one ML model. Detection only: no blocking, blackholing or AS_PATH analysis.

Background

Origin validation asks one question: is this AS allowed to announce this prefix?

Key concepts

- **AS:** an independently run network. The origin AS is the one that announces the prefix.
- **ROA:** a signed statement that an AS may announce a prefix, up to a maximum length (maxLength) [3].
- **ROV:** RFC 6811 labels a route Valid (a ROA matches), Invalid (a ROA covers it but none match) or NotFound (no ROA covers it) [1].
- **Operator inventory:** the origins, subprefixes and planned changes the network owner expects.

Monitoring approach

- Replay recorded BGP updates in the order they arrived.
- Check each update against the latest ROA snapshot taken on or before its timestamp.
- Apply two detection rules and the inventory, so known legitimate cases are suppressed.
- Raise an alert with its evidence. One ML model is added only after the baseline is measured.

Literature & RFC Review

Gap we address: ROV applied at the right time, with operator intent and evidence for every alert.

Standards

- **RFC 6480 [2]:** RPKI architecture and what its certificates mean.
- **RFC 9582 [3]:** format of a ROA (prefix, origin AS, optional maxLength).
- **RFC 6811 [1]:** the Valid, Invalid and NotFound states; our baseline check.
- **RFC 9319 [4]:** why a loose maxLength allows forged-origin subprefix hijacks.
- **RFC 4271 [5]:** BGP-4 UPDATE messages. We read only the prefix and origin.

Research, tools and data

- **BGPStream [6] and RIPE RIS [8]:** recorded updates that we replay in time order.
- **RIPE NCC RPKI archive [9]:** daily ROA snapshots, matched to update times.
- **ARTEMIS [7]:** operator-side hijack detection. A reference for our approach, not reused.

Proposed Solution

Replay RIS updates in time order, validate each against the matching ROA snapshot, then apply two rules.

- **R1, unexpected origin:** the origin AS is not allowed by any covering ROA or by the inventory entry for that prefix.
- **R2, more-specific:** the prefix is longer than the ROA maxLength, or is a subprefix the inventory does not list. This fires even when ROV says Valid [4].
- **Legitimate cases:** multi-homing, provider changes and DDoS mitigation are entered in the inventory with time windows. Matching alerts are downgraded and each suppression is logged.
- **Evidence:** each alert shows the prefix, observed and expected origin, the rule that fired, ROV state, covering ROA, snapshot date and reason.
- **Phase 2 ML:** a gradient-boosted model re-scores rule alerts and shows its top factors. We keep it only if it cuts false positives at the same recall.
- **Evaluation:** we inject origin, subprefix and forged-origin subprefix events, mix in legitimate cases, and report precision, recall, alert latency and false positives.

Optimization Proposed

Each optimization closes a gap in plain ROV, and we measure each one.

Optimization	Plain ROV	Our approach	How we measure
Time-aligned ROAs	Uses today's ROAs on old updates	Uses the snapshot from the update's date	Recall and false positives on ROA-change cases
Operator inventory	NotFound routes pass; a forged-origin subprefix can look Valid	Flags unexpected origins and unlisted subprefixes	Recall on forged-origin subprefix events
Legitimate-case suppression	Multi-origin and provider change look like hijacks	Inventory windows downgrade them; each is logged	False-positive rate on legitimate cases
ML re-scoring (Phase 2)	Fixed rule thresholds	Gradient-boosted re-ranking with top factors	Precision and false positives vs baseline at equal recall

Limits: ROA snapshots are daily, so same-day changes are missed. Forged-origin detection depends on how complete the inventory is. We do not analyse AS_PATH.

Solution Architecture & Design

Main components and how they connect; the ML scorer is an optional Phase 2 add-on.

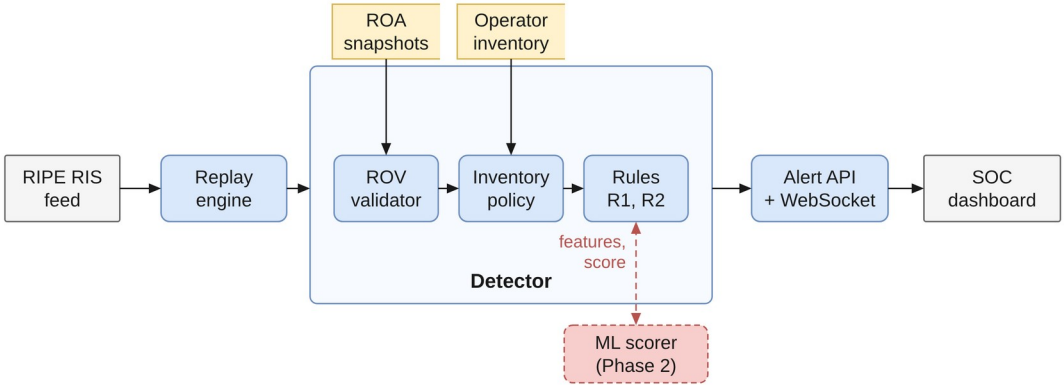


Figure: System Architecture UML Diagram

Data Flow Diagram

Level 1 DFD: from the RIPE sources to alerts and the evaluator.

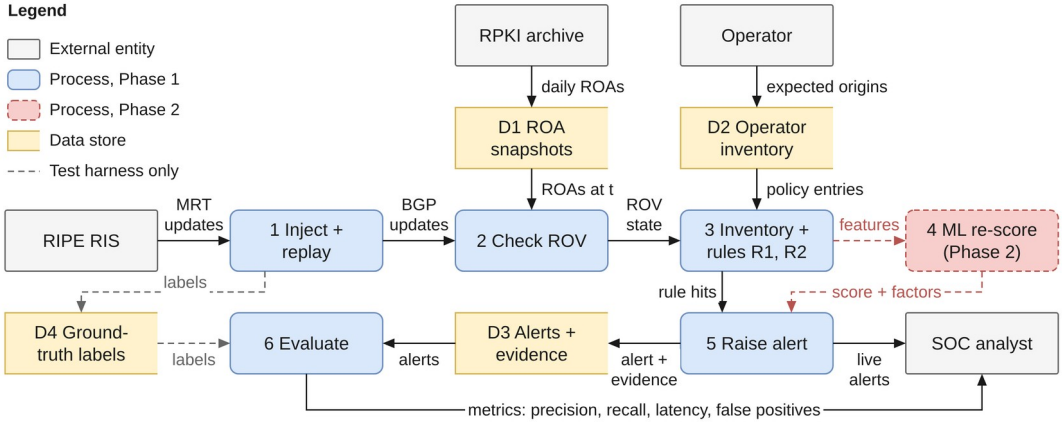


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Two weeks, four phases. ML starts only after the baseline has numbers.

Phase	Work	Exit check
1 Days 1-3	Data and labels: one RIS collector, a fixed replay window, matching ROA snapshots, a starter inventory and the anomaly injector.	Replay gives the same result every run; labels frozen.
2 Days 4-7	Rule baseline: ROV validator, inventory policy, rules R1 and R2, evidence alerts and the evaluation harness.	Baseline precision, recall, latency and false positives reported.
3 Days 8-11	ML re-scorer: gradient-boosted model with top factors, time-based split, unseen injection patterns held out.	Kept only if false positives drop at equal recall.
4 Days 12-14	Service and demo: FastAPI and WebSocket alert stream, SOC dashboard and final report.	Demo runs from a clean checkout with one command.

Method: fixed replay window and random seeds, injected labels kept separate from the detector, metrics defined before any tuning. If the baseline runs late, ML is dropped and the rule baseline is what we submit.

Bibliography I

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[9] RIPE NCC. "RPKI Repository Archive." https://ftp.ripe.net/rpki/

Thank You for Your Attention!